

FAANG Operating System Interview Master Sheet

How to Use

- Focus on concepts first, then examples, then numericals.
 - For every topic prepare: Definition, Why needed, Real-life example, Advantages, Limitations, Interview traps.
 - Practice explaining answers in 60 sec and 3 min versions.
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1. OS Fundamentals

1. What is an Operating System?
 2. Main functions of an OS.
 3. Types of Operating Systems.
 4. Kernel vs Operating System.
 5. Kernel mode vs User mode.
 6. What is a system call?
 7. What is booting? Cold boot vs Warm boot.
 8. Multiprogramming vs Multitasking vs Multiprocessing.
 9. Monolithic kernel vs Microkernel.
 10. What is a socket?
 11. Dynamic binding.
 12. Main memory vs Secondary memory.
 13. Cache memory and why it is needed.
 14. Direct mapping vs Associative mapping.
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2. Process & Thread Management

1. Process vs Program vs Thread.
2. Process states diagram.
3. Process Control Block (PCB).
4. Context switching and overhead.
5. Multithreading benefits.
6. User threads vs Kernel threads.
7. Zombie process.
8. Orphan process.
9. Daemon process.
10. Inter Process Communication (IPC).
11. Shared memory vs Message passing.
12. Why threads are lighter than processes.
13. Browser tabs: process or thread design?

3. CPU Scheduling (Must Prepare Numericals)

1. What is CPU scheduling?
2. Preemptive vs Non-preemptive scheduling.
3. FCFS Scheduling.
4. SJF Scheduling.
5. SRTF Scheduling.
6. LRTF Scheduling.
7. Priority Scheduling.
8. Round Robin Scheduling.
9. Time Quantum.
10. Throughput.
11. Turnaround Time.
12. Waiting Time.
13. Response Time.
14. Starvation.
15. Aging.
16. Compare FCFS vs SJF vs RR.
17. Design scheduler for gaming/video streaming.

4. Synchronization & Critical Section

1. What is synchronization?
 2. Critical Section Problem.
 3. Conditions of good critical section solution.
 4. Race condition.
 5. Mutex.
 6. Semaphore.
 7. Binary semaphore.
 8. Counting semaphore.
 9. Semaphore vs Mutex.
 10. Monitor.
 11. Spinlock.
 12. Busy waiting.
 13. Peterson's Solution.
 14. Producer Consumer Problem.
 15. Reader Writer Problem.
 16. Dining Philosophers Problem.
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5. Deadlock

1. What is deadlock?
 2. Four necessary conditions.
 3. Real-life deadlock example.
 4. Deadlock prevention.
 5. Deadlock avoidance.
 6. Deadlock detection.
 7. Deadlock recovery.
 8. Prevention vs Avoidance vs Detection.
 9. Resource Allocation Graph.
 10. Safe state vs Unsafe state.
 11. Deadlock in databases/servers.
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6. Banker's Algorithm

1. What is Banker's Algorithm?
 2. Why deadlock avoidance algorithm?
 3. Available matrix.
 4. Max matrix.
 5. Allocation matrix.
 6. Need matrix.
 7. How to find safe sequence.
 8. Unsafe state meaning.
 9. Numerical problems.
 10. Practical limitations.
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7. Memory Management

1. What is memory management?
2. Contiguous allocation.
3. Fragmentation.
4. Internal vs External fragmentation.
5. First Fit.
6. Best Fit.
7. Worst Fit.
8. Paging.
9. Why paging is needed.
10. Segmentation.
11. Paging vs Segmentation.
12. Virtual Memory.
13. Demand Paging.

14. Thrashing (why it occurs).
 15. Swapping.
 16. TLB.
 17. Belady's Anomaly.
 18. Working Set Model.
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8. Page Replacement Algorithms (Must Prepare Numericals)

1. Page fault.
 2. FIFO.
 3. LRU.
 4. Optimal.
 5. Second Chance / Clock.
 6. FIFO vs LRU.
 7. Why LRU better than FIFO.
 8. Belady's anomaly explanation.
 9. Solve page reference string problems.
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9. File Systems & Disk Management

1. What is file system?
 2. File access methods.
 3. Directory structure.
 4. Single-level vs Two-level directory.
 5. Inode.
 6. File allocation methods.
 7. Contiguous allocation.
 8. Linked allocation.
 9. Indexed allocation.
 10. Free space management.
 11. What is RAID?
 12. RAID levels.
 13. Disk scheduling.
 14. FCFS disk scheduling.
 15. SSTF.
 16. SCAN.
 17. C-SCAN.
 18. SSD vs HDD performance.
 19. What is spooling?
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10. Security & Protection

1. Protection vs Security.
 2. Authentication vs Authorization.
 3. Access control matrix.
 4. Malware.
 5. Virus vs Worm vs Trojan.
 6. Privilege escalation.
 7. How OS provides security.
 8. Process isolation.
 9. Permissions in Linux.
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11. Linux / System Calls (Interview Favorite)

1. What are system calls?
 2. fork() vs exec().
 3. wait().
 4. pipe().
 5. open(), read(), write(), close().
 6. What is shell?
 7. What is kernel panic?
 8. Basic Linux commands.
 9. ps, top, kill, grep, chmod, ls, pwd.
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12. FAANG Scenario-Based Questions

1. How does OS run many apps together?
 2. Why does system slow when RAM is full?
 3. Why are threads preferred in web servers?
 4. How does browser open multiple tabs efficiently?
 5. How would you prevent deadlock in banking server?
 6. Why does SSD outperform HDD?
 7. How does OS recover after crash?
 8. How to design scheduler for video streaming app?
 9. Why context switching hurts performance?
 10. Why too many threads can reduce speed?
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13. Top 25 Quick Revision Questions

1. Process vs Thread.

2. Program vs Process.
 3. Context Switching.
 4. PCB.
 5. Deadlock conditions.
 6. FCFS vs SJF vs RR.
 7. Starvation vs Aging.
 8. Paging vs Segmentation.
 9. Virtual Memory.
 10. Thrashing.
 11. FIFO vs LRU.
 12. Belady's Anomaly.
 13. Safe vs Unsafe state.
 14. fork() vs exec().
 15. Semaphore vs Mutex.
 16. Binary semaphore.
 17. Monolithic vs Microkernel.
 18. TLB.
 19. RAID.
 20. Spooling.
 21. Cache memory.
 22. Internal vs External fragmentation.
 23. Demand paging.
 24. Zombie vs Orphan process.
 25. User mode vs Kernel mode.
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Interview Strategy

- Start with definition.
- Explain why needed.
- Give real-world example.
- Mention trade-offs.
- End with practical use case.
- Draw diagrams in live interviews when possible.